

Path Planning Techniques for Multi-robot Systems: A Systematic Review

Nour Ayman Abujabal

Department of Computer Engineering
University of Sharjah
Sharjah, United Arab Emirates
U21103451@sharjah.ac.ae

Tamer Rabie

Department of Computer Engineering
University of Sharjah
Sharjah, United Arab Emirates
trabie@sharjah.ac.ae

Ibrahim Kamel

Department of Computer Engineering
University of Sharjah
Sharjah, United Arab Emirates
kamel@sharjah.ac.ae

Abstract—Numerous path planning studies have been conducted in past decades due to the challenges of obtaining optimal solutions. This paper provides a comprehensive review of multi-robot path planning approaches, focusing on classical, AI-based, and hybrid methods. Bio-inspired techniques are commonly used, and AI approaches are gaining popularity. The challenges faced by multi-robot systems include the number of robots, energy efficiency, fault tolerance, robustness, and dynamic targets. Therefore, developers are continuously researching for optimal solutions that can gather the benefits of multi-robot systems while overcoming the possible challenges. This paper provides a comprehensive assessment of various path planning techniques, highlights key problems in this field, and identifies research gaps for a better multi-robot path planning experience.

Index Terms—Path planning, multi-robot, classical approaches, artificial intelligence approaches, hybrid approaches.

I. INTRODUCTION

Recent years have seen a sharp rise in the demand for robots, which are now used extensively in all fields and industries. Multi-robot systems are becoming increasingly important in a variety of fields, including manufacturing, transportation, and healthcare. Implementing a multi-robot system, which consists of multiple robots working together, has several benefits. One of the advantages is that multi-robot systems have the potential to increase the efficiency and effectiveness of tasks that would be difficult or impossible for a single robot to accomplish [1]. For example, search and rescue missions can be improved by using teams of robots that can search a larger area more quickly and effectively than a single robot [2]. Another benefit is that the problem of a single robot failure is mitigated in a multi-robot system, as other robots can take over the mission, thus making the system more robust.

Multi-robot systems can be composed of homogeneous systems or heterogeneous systems. The main difference between homogeneous and heterogeneous systems in robotics is the capabilities and functions of the robots within the system [3]. Homogeneous systems consist of identical robots that are capable of performing the same tasks, while heterogeneous systems consist of robots with different capabilities and functions. Some examples of homogeneous systems include: Unmanned Ground Vehicles (UGVs), Unmanned Aerial Vehicles (UAVs), and Unmanned Surface Vehicles (USVs). Examples of heterogeneous systems include a combination

of UAVs and UGVs (UAV-UGVs), USVs with Unmanned Underwater Vehicles (USV-UUVs), UAVs with Manipulators (Aerial Manipulators), and UGVs with Manipulators (Mobile Manipulators).

This paper aims to classify path planning techniques based on various factors such as obstacles, destination, communication, shortest time, and shortest path. The classification will consider the nature of obstacles encountered during path planning, distinguishing between static, dynamic, and obstacle-free scenarios. Furthermore, it will examine whether the destination remains the same or differs from the starting point, and whether the destination itself is static or dynamic. Another important aspect being considered is the presence of communication capabilities during the path planning process. Additionally, the paper will explore whether the objective is to find the shortest time or the shortest path for reaching the destination. By categorizing path planning techniques based on these criteria, this study aims to provide a comprehensive understanding of the different approaches used in various scenarios, aiding in the selection of the most appropriate technique for specific application requirements.

The rest of the paper is organized as follows: In Section II, a brief background of path planning and relevant research is provided. Section III outlines the approach used for conducting this review. Section IV offers a comprehensive taxonomy and reviews various methods within each category. In Section V, the main findings from evaluating mechanisms in the literature are summarized, and the existing challenges in multi-robot systems path planning are discussed. Finally, Section VI presents the concluding remarks.

II. BACKGROUND AND RELATED WORK

Path planning is a fundamental problem in robotics and autonomous systems, involving the determination of an optimal path for a robot to navigate from a starting point to a destination while considering obstacles and objectives like minimizing travel time or distance. Numerous endeavors have been made to create a comprehensive review of path planning methods for multi-robot systems operating in various environments. However, recent literature reviews have been constrained to specific robot types, such as UGV systems [4]–[7], and UAV systems [8]. Moreover, another review neglects

communication which is an essential process while dealing with multi-robot systems [9].

In [4], the authors conducted a thorough review of path planning techniques for autonomous mobile robots, categorizing them into Soft-Computing, Reactive-Computing, C-Space Search, and Optimal Control Algorithms. They considered important characteristics for each category, including preliminary map model, deterministic, replanning, optimality, and completeness. Meanwhile, the authors in [5] presented different categorizations of path planning, considering both global and local approaches, as well as examining both classical and heuristic methods. The authors of [10] presented an overview of trajectory planning algorithms, categorizing them into bio-inspired, metaheuristic, sampling-based, and decomposition graph-based approaches and comparing them based on several criteria. Furthermore, [7] discussed multi-robot path planning and decision-making, categorizing approaches as classical, heuristic, bio-inspired, and AI-based while considering centralized and decentralized decision-making strategies. Lastly, [11] conducted a detailed study on multi-robot systems coordination, covering basic terminology, classification, and different application aspects. They provided valuable perspectives on coordination methods for each aspect, analyzed strengths, limitations, open research issues, and highlighted future work opportunities.

To the best of the author's knowledge, this paper represents a pioneering effort in its domain, as the proposed classification framework, which considers factors such as obstacle type, destination characteristics, communication availability, and optimization objectives, has not been previously addressed. While path planning techniques have been extensively studied, a comprehensive categorization and analysis of these techniques based on the aforementioned criteria remains unexplored. By filling this gap in the existing literature, this study aims to provide valuable insights and establish a foundation for future research in the field of path planning and optimization.

III. METHODOLOGY

This paper conducts a comprehensive review following Kitchenham and Charters's Systematic Literature Review (SLR) guidelines [12]. The methodology employed here encompasses three main phases: planning, conducting, and reporting. The scope of the search is limited to literature published between 2018 and 2023, specifically focusing on path planning techniques for multi-robot systems. By following the search protocol, the review includes a total of 46 articles, with an annual distribution of these articles shown in Figure 1.

In the remaining part of this section, crucial details about the methodology are provided. Initially, research questions and the search protocol are established in the planning phase. Subsequently, the study selection and exclusion criteria are presented during the conducting phase. Finally, data extraction and result reporting are conducted.

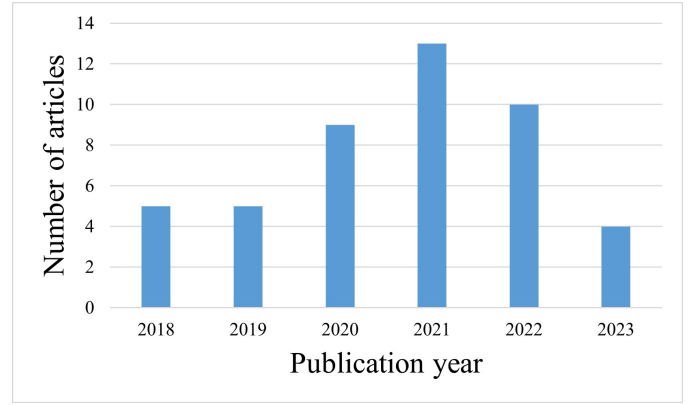


Fig. 1. The number of included publications per year.

A. Planning Phase - Research Questions

The purpose of this study is to address the following questions:

RQ1: What are the most used path planning approaches for multi-robot systems?

RQ2: Are Hybrid path planning techniques improve the traditional approach in terms of time, path length, and smoothness?

RQ3: What are the main challenges of multi-robot systems that can affect the factors of path planning?

B. Conducting Phase - Search Strategy

In this SLR, the search strategy was designed in accordance with the review's objectives and research questions. The search process consists of three stages: a term-based search, a crawling-based search, and the application of inclusion/exclusion criteria.

1) *Search Terms*: Based on the the classes of path planning techniques that are shown in Tables I the papers are collected using the main key search terms word including "path planning" AND "dynamic obstacles" OR "static obstacle" AND "multiple robots" OR "multi-robot".

2) *Study Selection*: Using the stated search strategy, 230 publications were acquired. However, to align with the survey's objectives, further filtration was required as many were off-topic. Thus, specific inclusion/exclusion rules were applied.

- Selection Criteria:
 - Papers from 2018 to 2023.
 - Papers of path planning techniques for multi-robot systems.
 - Papers for static and dynamic Obstacles.
 - Papers presenting same and different destinations.
 - Papers of recent bio-inspired path planning techniques.
- Exclusion Criteria:
 - Duplicated papers.
 - Papers related to single robot system.
 - Rapers related to control.

TABLE I
SUMMARY OF PATH PLANNING WORKS IN THE LITERATURE.

Class	Technique	Paper	Year	Obstacles (Static/Dynamic/ Obstacle free)	Destination (Same/Different)	Destination (Static/Dynamic)	Communication (Yes/No)	Shortest time (Yes/No)	Shortest path (Yes/No)
Grid-based techniques	A*	[13] [14] [15]	2021 2020 2023	[13] Static [14], [15] Dynamic	Different	Static	[13] Yes [14], [15] No	[13], [14] No [15] Yes	Yes
	Theta*	[16]	2022	Dynamic	Different	Static	Yes	Yes	Yes
	Greedy algorithms	[17]	2020	Dynamic	Different	Static	Yes	No	No
	Dijkstra	[18]	2021	Dynamic	Different	Static	No	No	Yes
Sampling-based techniques	PRM	[19]	2023	Dynamic	Different	Static	Yes	Yes	No
	RRT	[20]	2018	Static	Same	Static	Yes	Yes	No
	RRT*	[21]	2021	Dynamic	Same	Static	No	Yes	Yes
APF	-	[22] [23]	2022 2022	[22] Obstacle free [23] Static	[22] Different [23] Same	Static	Yes	[22] No [23] Yes	Yes
	RNN	[24]	2021	Dynamic	Same	Static	Yes	No	No
Artificial neural network	Adaptive neural network	[25]	2018	Static	Same	Dynamic	Yes	Yes	Yes
	Globally guided RL	[26]	2020	Dynamic	Same	Static	No	No	Yes
Reinforcement learning	Deep Q-learning	[27]	2021	Static	Different	Static	Yes	Yes	No
Bio-Inspired algorithms	GA	[28] [29]	2019 2020	[28] Static [29] Dynamic	Different	Static	[28] No [29] Yes	No	Yes
	BFO	[30]	2020	Dynamic	Same	Static	No	No	Yes
	PSO	[31]	2021	Dynamic	Different	Static	No	No	Yes
	Ant-lion optimization	[32]	2020	Dynamic	Different	Static	Yes	No	Yes
	ACO	[33]	2022	Static	Same	Static	Yes	Yes	Yes
	Grasshopper optimization	[34]	2021	Dynamic	Same	Both	No	Yes	Yes
Hybrid techniques	Classical-optimization	[35] [36] [37] [38]	2022 2022 2022 2021	[35] Obstacle free [36], [37] Dynamic [38] Static	[36], [38] Same [37], [35] Different	[36] Dynamic [35], [37], [38] Static	[35], [37], [36] No [38] Yes	[35] Yes [36], [37], [38] No	[35], [36], [38] Yes [37] No
	Multiple-classical	[39]	2018	Dynamic	Same	Static	Yes	No	Yes
	Multiple-optimization	[40] [41] [42]	2022 2018 2023 2023	Dynamic	[40], Same [41], [42] Different	Static	[40], [42] No [41] Yes	Yes	Yes

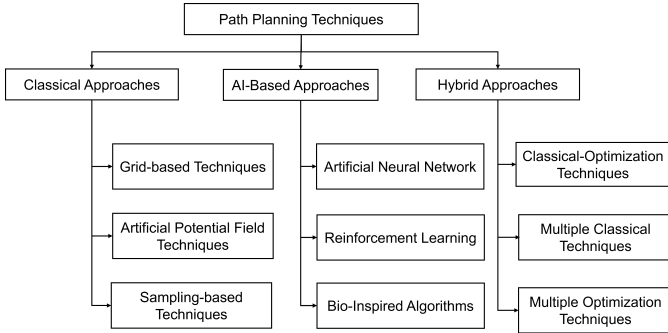


Fig. 2. Classification of path planning techniques for multi-robot systems.

IV. PATH PLANNING CLASSIFICATION

Path planning approaches can primarily be categorized into three groups: classical approaches, Artificial Intelligence (AI)-based approaches, and hybrid approaches, as depicted in Figure 2. The following subsections provide a concise overview of each category and presents a summary of various studies mentioned in the literature. Table I summarizes most of the works done on path planning for multi-robot systems in the past six years.

A. Classical Approaches

Before the rise of AI approaches, classical path planning approaches were extensively utilized for robot navigation. Variety classical path planning approaches were conducted including grid-based [13]–[18], artificial potential field [22], [23], and sampling-based techniques [20], [21]. The following subsections discuss the three subcategories of classical approaches.

1) Grid-based Techniques: Grid-based algorithms are classical path planning techniques that divide the environment into a grid of cells. There are four grid-based techniques including A*, Dijkstra, Theta*, and Greedy algorithms. A* is a widely used path planning algorithm that efficiently finds the shortest path in a graph or grid. A* ensures completeness, meaning it can find the best possible solution if one exists. Similar to A*, Dijkstra algorithm aims to discover the optimal path within a graph but with known edge weights. Unlike A* algorithm, Dijkstra algorithm must exhaustively search the entire graph to reach the solution. Therefore, A* achieves improved results when dealing with problems that involve large environments. Several works utilized A* such as [13]–[15]. Theta* is an extension of the A*, incorporating additional improvements to enhance efficiency. Unlike A*, Theta* allows the parent of a node to be any node, not necessarily its direct

neighbor, as long as there is a clear line-of-sight between the two nodes. As a result, Theta* reduces the number of nodes to evaluate by considering the visibility between successive nodes [16]. Lastly, Greedy algorithm is a simple yet effective approach used in path planning to find a feasible solution. It focuses on making locally optimal decisions at each step of the process, without considering the overall path or the potential consequences of those choices. It involves the iterative division of a set of resources, prioritizing the maximum immediate availability of each resource at each stage of execution [17].

2) *Artificial Potential Field (APF) Techniques*: APF algorithms are a type of classical path planning techniques that simulate forces in a virtual potential field to guide the movement of robots. These algorithms create attractive forces towards the goal location and repulsive forces from obstacles. APF algorithms can be used in scenarios where the environment is known and obstacles can be detected. The main advantage of APF algorithm lies in its quick convergence, as it can achieve the final path with fewer computations compared to grid-based and sampling-based methods. Additionally, another advantage is its ability to handle dynamic environments, as the forces can be updated in real time. They can also provide smooth and continuous paths to the goal. However, APF algorithms have limitations. They can get trapped in local minima or maxima of the potential field. Additionally, it can be challenging to tune the parameters of the potential field to ensure a balance between avoiding obstacles and reaching the goal efficiently. APF technique has been implemented in several works in the literature [22], [23].

3) *Sampling-based Techniques*: Sampling-based techniques are a class of classical path planning algorithms that rely on random sampling to explore and navigate through an environment. There are several sampling-based techniques including Rapidly Exploring Random Tree (RRT) [20], RRT* [21], and Probabilistic Roadmap (PRM) [19] algorithms. RRT and PRM algorithms are popular path planning algorithms which aim to sample the environment while considering the starting and ending positions. RRT algorithm incrementally constructs a tree-like structure by randomly sampling points in the configuration space and connecting them to the existing tree. The expansion of the tree is biased towards unexplored areas, allowing for rapid exploration of the search space. In addition, RRT excels in scenarios where the exact solution is not required, as it focuses on quickly generating feasible paths. On the other hand, RRT* improves upon RRT by adding a rewiring step that optimizes the tree structure by considering nearby nodes. This rewiring step allows RRT* to continuously improve and refine the paths it generates, converging towards an optimal solution over time.

B. AI-based Approaches

AI path planning approaches have emerged as a significant improvement over classical approaches. Unlike classical methods that rely on predetermined mathematical models and algorithms, AI-based path planning leverages the power of machine learning and other AI techniques. This enables the

algorithms to learn from data and adapt to dynamic and unpredictable environments. The following subsections discuss the three subcategories of AI-based approaches.

1) *Artificial Neural Network (ANN)*: ANN is a machine learning method that employs neural networks to find optimal paths for robots. ANN-based systems are highly valuable for mobile robotic navigation due to their characteristics, such as generalization, distributed representation, parallelism, fault tolerance, and learning ability [25]. Nonetheless, a drawback of ANN-based techniques is their time-consuming nature, and the learning process may not always guarantee convergence to the optimal solution.

2) *Reinforcement Learning (RL)*: RL is a technique used in path planning where an agent learns to make sequential decisions to maximize a cumulative reward. In RL-based path planning, the agent interacts with the environment, receives feedback in the form of rewards, and adjusts its actions based on these rewards. RL is particularly useful in scenarios where the environment is unknown, complex, or dynamic. It allows the agent to learn optimal policies through exploration and exploitation. There are several types of reinforcement learning techniques commonly used in path planning, such as Q-learning and Deep Q-Networks (DQN).

3) *Bio-Inspired Algorithms*: Bio-inspired techniques, commonly referred to as biomimicry, encompass a range of methodologies that draw inspiration from the natural world. These techniques seek to imitate the structures, behaviors, and functionalities observed in various natural systems, including the human body, animals, and plants [43]. There are several types of bio-inspired algorithms in the literature including Genetic Algorithms (GA) [28], [29], Particle Swarm Optimization (PSO) [31], Ant Colony Optimization (ACO) [33], Bacterial Foraging Optimization (BFO) [30], Ant Lion Optimization (ALO) [32], and Grasshopper Algorithm [34].

C. Hybrid Approaches

Hybrid path planning techniques combine multiple approaches or methodologies to achieve optimal path planning solutions. In this paper, the hybrid techniques are divided into three main categories which are classical-optimization, multiple classical, and multiple optimization techniques. Several works in the literature combined different path planning techniques and developed hybrid techniques. Various works used different techniques with Grid-based algorithms, such as Dijkstra algorithm and Simulated Annealing (SA) approach [35], A* and Potential Field [39], and A* and Reinforcement Learning [37]. Other works focused on combining different optimization techniques to handle the challenges in path planning, such as Wolf Swarm Algorithm and Artificial Potential Field (WSA-APF) [36], Modified hyperbolic Gravitational Search Algorithm and Dynamic Window Approach (MGSA-DWA) [41], Self Organizing Migrating Algorithm and Particle Swarm Optimization (SOMA-PSO) [42], and Grey Wolf Optimizer and Whale Optimizer Algorithm (GWO-WOA) [40]. Finally, the combination of sampling-based and optimization

algorithms has been employed in [38] in order to solve the path planning problem.

V. RESULTS AND DISCUSSION

Reaching this point, this work has reviewed more than 40 studies that are concerned with path planning for multi-robot systems. The reason for the increasing number of studies on path planning in a multi-robot system is that it is a fascinating and challenging field that holds immense potential for revolutionizing various domains. The motivation for exploring path planning in multi-robot systems stems from the promise of enhanced coordination, increased efficiency, and improved overall system performance. The synergy of multiple robots working together, guided by intelligent path planning strategies, can lead to faster and more accurate completion of tasks, improved productivity, and increased scalability. The remaining of this section illustrates some observations about the main strengths and limitations of various path planning approaches used in the literature. Finally, the challenges and gaps in path planning for multi-robot systems will be addressed.

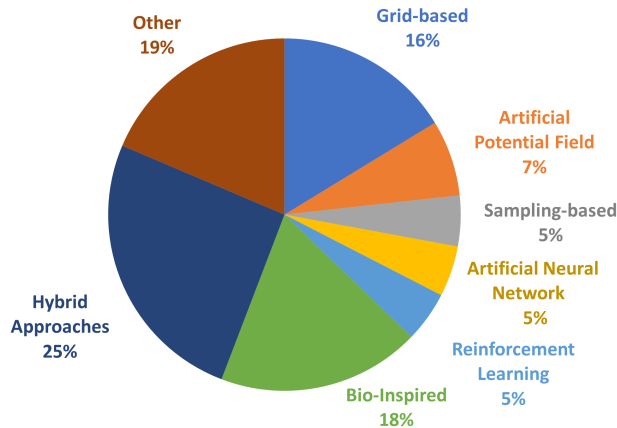


Fig. 3. The percentage of articles under each category of path planning.

A. Methods Evaluation

Selecting the appropriate approaches is a vital decision in system development. Given that one of the primary aims of this review is to provide guidance for future research in this domain, this section will concisely discuss the strengths and weaknesses of the main methods used in the reviewed literature. Figure 3 shows the percentage of articles under each category of path planning.

Based on the findings in [23], it is proved from the performance evaluation metrics of path length and travel time that the proposed APF algorithm outperforms existing GA, RRT, and A* algorithms. In [28], a comparative analysis between the proposed EGA and A*, PRM, B-RRT, and PSO demonstrated the superiority of the proposed EGA in terms of path length, runtime, path smoothness, and success rate across 12 environments of varying sizes and complexities. The study revealed a consistent 100% success rate for the proposed

EGA, irrespective of the environment's size and complexity. In [34], the Grasshopper-based technique was compared with GA, PSO, Neuro-Fuzzy, D*, and RRT*. The simulation results showcased the effectiveness of the proposed technique in guiding the robot to the target, avoiding collisions, and finding the shortest path in less time when compared to the other techniques. The authors in [27] compared the DQN algorithm to other 2D baseline algorithms which are RRT, RRT*, and A* in terms of computational time. The results show that, on average, the DQN had the lowest computation time and was comparable to A* in terms of its path length to the goal.

B. Research Gaps

As aforementioned, several challenges have been identified that impede the progress of research in the domain of path planning for multi-robot systems. These challenges include types of robotic systems, number of robots in the system, energy efficiency, fault tolerance and robustness, considered assumptions, shape of obstacles, and dynamic targets. This section aims to identify and discuss certain gaps present in the current literature.

The literature primarily focuses on path planning approaches for homogeneous systems, with limited attention to heterogeneous multi-robot systems. Furthermore, increasing the number of robots in the system poses a challenge as it affects the proposed technique's performance and efficiency in finding the optimal path. Other factors that must be considered while deploying multi-robot systems are robustness and energy efficiency which are neglected in the literature. There are different situations that could degrade the robustness of a multi-robot system, such as communication failures, sensors malfunctions, and leader failure. Moreover, different shapes of the obstacles are not considered in the majority of the work in the literature. Finally, only a few works proposed a single dynamic target, such as [25], [34], [36]. Additionally, it will be more challenging to deal with multi-dynamic targets.

VI. CONCLUSION

This paper conducted a systematic review of recent path planning approaches for multi-robot systems published from 2018 to 2023. It started with an introduction to path planning techniques and identified current research directions while pinpointing gaps in the research workflow. The study then categorized the latest path planning techniques based on factors such as obstacles, destination, communication, shortest time, and shortest path. Moreover, it highlighted a lack of literature on heterogeneous systems and dynamic target path planning. Addressing these research gaps is crucial for future advancements in the field of multi-robot systems' path planning.

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